
Statistical Inference

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Exercise Series 10

Exercise 1. In the following, under different setups, we provide a statistical model, a hypothesis, and three different tests φ_{left} , φ_{right} and φ_{both} , called the *left-sided*, the *right-sided*, and the *two-sided* or *both-sided* tests. In each case, find the power function of the three tests. Determine the constants c_{left} , c_{right} , c_1 and c_2 , so that all three tests have the same size (say α). Draw the power functions to determine which of the three tests is the most appropriate against the alternatives $\mathcal{H}_{\text{left}}$, $\mathcal{H}_{\text{right}}$ and $\mathcal{H}_{\text{both}}$, respectively.

(a) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$, $0 \leq p \leq 1$. $\mathcal{H} : p = \frac{1}{4}$. $\mathcal{H}_{\text{left}} : p < \frac{1}{4}$, $\mathcal{H}_{\text{right}} : p > \frac{1}{4}$, $\mathcal{H}_{\text{both}} : p \neq \frac{1}{4}$.

$$\begin{aligned}\varphi_{\text{left}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i < c_{\text{left}}, \\ 0 & \text{otherwise,} \end{cases} & \varphi_{\text{right}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i > c_{\text{right}}, \\ 0 & \text{otherwise,} \end{cases} \\ \varphi_{\text{both}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i < c_1 \text{ or } \sum_{i=1}^n x_i > c_2, \\ 0 & \text{otherwise.} \end{cases}\end{aligned}$$

(b) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$, $\lambda > 0$. $\mathcal{H} : \lambda = 2$. $\mathcal{H}_{\text{left}} : \lambda < 2$, $\mathcal{H}_{\text{right}} : \lambda > 2$, $\mathcal{H}_{\text{both}} : \lambda \neq 2$.

$$\begin{aligned}\varphi_{\text{left}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i < c_{\text{left}}, \\ 0 & \text{otherwise,} \end{cases} & \varphi_{\text{right}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i > c_{\text{right}}, \\ 0 & \text{otherwise,} \end{cases} \\ \varphi_{\text{both}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i < c_1 \text{ or } \sum_{i=1}^n x_i > c_2, \\ 0 & \text{otherwise.} \end{cases}\end{aligned}$$

(c) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(0, \theta)$, $\theta > 0$. $\mathcal{H} : \theta = 1$. $\mathcal{H}_{\text{left}} : \theta < 1$, $\mathcal{H}_{\text{right}} : \theta > 1$, $\mathcal{H}_{\text{both}} : \theta \neq 1$.

$$\begin{aligned}\varphi_{\text{left}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } x_1 < c_{\text{left}}, \\ 0 & \text{otherwise,} \end{cases} & \varphi_{\text{right}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } x_1 > c_{\text{right}}, \\ 0 & \text{otherwise,} \end{cases} \\ \varphi_{\text{both}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } x_1 < c_1 \text{ or } x_1 > c_2, \\ 0 & \text{otherwise.} \end{cases}\end{aligned}$$

(d) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu, 1)$, $\mu \in \mathbb{R}$. $\mathcal{H} : \mu = 5$. $\mathcal{H}_{\text{left}} : \mu < 5$, $\mathcal{H}_{\text{right}} : \mu > 5$, $\mathcal{H}_{\text{both}} : \mu \neq 5$.

$$\begin{aligned}\varphi_{\text{left}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } x_1 + x_2 < c_{\text{left}}, \\ 0 & \text{otherwise,} \end{cases} & \varphi_{\text{right}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } x_1 + x_2 > c_{\text{right}}, \\ 0 & \text{otherwise,} \end{cases} \\ \varphi_{\text{both}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } x_1 + x_2 < c_1 \text{ or } x_1 + x_2 > c_2, \\ 0 & \text{otherwise.} \end{cases}\end{aligned}$$

(e) $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(0, \sigma^2)$, $\sigma > 0$. $\mathcal{H} : \sigma = 10$. $\mathcal{H}_{\text{left}} : \sigma < 10$, $\mathcal{H}_{\text{right}} : \sigma > 10$, $\mathcal{H}_{\text{both}} : \sigma \neq 10$.

$$\begin{aligned}\varphi_{\text{left}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i < c_{\text{left}}, \\ 0 & \text{otherwise,} \end{cases} & \varphi_{\text{right}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i > c_{\text{right}}, \\ 0 & \text{otherwise,} \end{cases} \\ \varphi_{\text{both}}(\mathbf{x}) &= \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i < c_1 \text{ or } \sum_{i=1}^n x_i > c_2, \\ 0 & \text{otherwise.} \end{cases}\end{aligned}$$

Hint: You can use the R functions starting with `p` to calculate the cdf of standard distributions. For example, `pbinom` gives you the cdf of Binomial distribution, `pnorm` gives you the cdf of Normal distribution, and so on.

Hint: To compare different tests, you should ideally plot the power functions on the same graph. At least, make sure that the axes of the different plots are the same, so that you don't get misguided by what you see.

Exercise 2. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Normal}(\mu, 1)$, $\mu \in \mathbb{R}$. Suppose that we are interested in testing $\mathcal{H}_0 : \mu = 5$ against $\mathcal{H}_a : \mu > 5$. Consider the following two tests for this problem.

$$\varphi_1(\mathbf{x}) = \begin{cases} 1 & \text{if } x_1 > c_1 \\ 0 & \text{otherwise,} \end{cases} \quad \varphi_2(\mathbf{x}) = \begin{cases} 1 & \text{if } \sum_{i=1}^n x_i > c_2 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the power functions of the two tests φ_1 and φ_2 .
- (b) Determine c_1 and c_2 so that the two tests have the same size.
- (c) Which of these two tests is most appropriate for testing $\mathcal{H}_0$ against $\mathcal{H}_a$?

Exercise 3. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Uniform}(0, \theta)$, $\theta > 0$. Suppose that we are interested in testing $\mathcal{H}_0 : \theta = 5$ against $\mathcal{H}_a : \theta > 5$. Consider the following two tests for this problem.

$$\varphi_1(\mathbf{x}) = \begin{cases} 1 & \text{if } x_1 > c_1 \\ 0 & \text{otherwise,} \end{cases} \quad \varphi_2(\mathbf{x}) = \begin{cases} 1 & \text{if } x_{(n)} > c_2 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the power functions of the two tests φ_1 and φ_2 .
- (b) Determine c_1 and c_2 so that the two tests have the same size.
- (c) Which of these two tests is most appropriate for testing $\mathcal{H}_0$ against $\mathcal{H}_a$?